

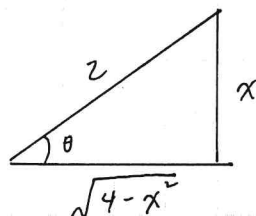
10901 微積分乙期末考評分標準

題號 1 (1)

題目: $\int \sqrt{16-4x^2} dx$.

let $x = 2 \sin \theta$

$dx = 2 \cos \theta d\theta$.



寫出正確假設 (得 2 分)

$\Rightarrow \int \sqrt{16-4x^2} dx = 2 \int \sqrt{4-x^2} dx$

$= 2 \int 2 \cos \theta \cdot 2 \cos \theta d\theta = 8 \int \cos^2 \theta d\theta$

寫出此步驟 (得 2 分)

$= 8 \int \frac{1 + \cos(2\theta)}{2} d\theta$

寫出此步驟 (得 2 分)

$= 4 \left(\theta + \frac{1}{2} \sin(2\theta) \right) + C$

$= 4 \left(\sin^{-1}\left(\frac{x}{2}\right) + \frac{1}{2} \left(2 \cdot \frac{x}{2} \cdot \frac{\sqrt{4-x^2}}{2} \right) \right) + C$

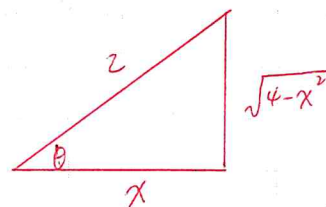
$= 4 \sin^{-1}\left(\frac{x}{2}\right) + x \sqrt{4-x^2} + C$

正確答案 (得 2 分)

若假設 $x = 2 \cos \theta$

$dx = -2 \sin \theta$

則解答為 $-4 \cos^{-1}\left(\frac{x}{2}\right) + x \sqrt{4-x^2} + C$



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題號 1 (2)

$$\int_1^{\infty} \frac{1}{x(\ln x)^3} dx = \int_1^2 \frac{1}{x(\ln x)^3} dx + \int_2^{\infty} \frac{1}{x(\ln x)^3} dx$$

合計3分

$$= \lim_{\Delta \rightarrow 1^+} \int_{\Delta}^2 \frac{1}{x(\ln x)^3} dx + \lim_{t \rightarrow \infty} \int_2^t \frac{1}{x(\ln x)^3} dx$$

(2分)

$$= \lim_{\Delta \rightarrow 1^+} \left(\frac{-1}{2(\ln x)^2} \right)_{\Delta}^2 + \lim_{t \rightarrow \infty} \left(\frac{-1}{2(\ln x)^2} \right)_2^t$$

(1分)

$$= \lim_{\Delta \rightarrow 1^+} \left(\frac{1}{2(\ln \Delta)^2} - \frac{1}{2(\ln 2)^2} \right) + \lim_{t \rightarrow \infty} \left(\frac{1}{2(\ln 2)^2} - \frac{1}{2(\ln t)^2} \right)$$

(2分)

$$= \infty$$

* $\int \frac{1}{x(\ln x)^3} dx = -\frac{1}{2}(\ln x)^{-2} + C$

只要把此部份積分算出來,即可得2分!

* $\Delta \rightarrow 1^+$ 的地方

若是寫 $\Delta \rightarrow 1$, 則得1分!!!

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題號 1 (3)

過程: $\int x^4 \cos(2x) dx$

$$\begin{array}{rcl}
 x^4 & \searrow & \cos(2x) \\
 -4x^3 & \searrow & \frac{1}{2} \sin(2x) \\
 12x^2 & \searrow & -\frac{1}{4} \cos(2x) \\
 -24x & \searrow & -\frac{1}{8} \sin(2x) \\
 24 & \searrow & \frac{1}{16} \cos(2x) \\
 0 & \searrow & \frac{1}{32} \sin(2x)
 \end{array}$$

or

$$\begin{aligned}
 & \int x^4 \cos(2x) dx \\
 &= \frac{1}{2} x^4 \sin(2x) - 2 \int x^3 \sin(2x) dx \\
 &= \frac{1}{2} x^4 \sin(2x) + x^3 \cos(2x) - 3 \int x^2 \cos(2x) dx \\
 &= \frac{1}{2} x^4 \sin(2x) + x^3 \cos(2x) - \frac{3}{2} x^2 \sin(2x) \\
 &\quad + 3 \int x \sin(2x) dx \\
 &= \frac{1}{2} x^4 \sin(2x) + x^3 \cos(2x) - \frac{3}{2} x^2 \sin(2x) \\
 &\quad - \frac{3}{2} x \cos(2x) + \frac{3}{2} \int \cos(2x) dx \\
 &= \frac{1}{2} x^4 \sin(2x) + x^3 \cos(2x) - \frac{3}{2} x^2 \sin(2x) \\
 &\quad - \frac{3}{2} x \cos(2x) + \frac{3}{4} \sin(2x) + C
 \end{aligned}$$

評分順序:

Step 1: 沒用分部積分, 直接 0 分 (沒有計算過程亦 0 分)

Step 2: $\frac{1}{2} x^4 \sin(2x) + x^3 \cos(2x) - \frac{3}{2} x^2 \sin(2x) - \frac{3}{2} x \cos(2x) + \frac{3}{4} \sin(2x) + C$

1pt
1pt
1pt
1pt
1pt
3pts

Step 3: 多寫的 扣 1pt

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題號 1. (4)

(法一) 微 積

$$\begin{array}{r} \sin(7x) \oplus \cos(3x) \\ 7 \cos(7x) \ominus \frac{1}{3} \sin(3x) \\ -49 \sin(7x) \oplus -\frac{1}{9} \cos(3x) \end{array}$$

(2)

or 積 微

$$\begin{array}{r} \sin(7x) \oplus \cos(3x) \\ -\frac{1}{7} \cos(7x) \ominus -3 \sin(3x) \\ -\frac{1}{49} \sin(7x) \oplus -9 \cos(3x) \end{array}$$

(2)

$$\begin{aligned} & \int \sin(7x) \cos(3x) dx \\ &= \frac{1}{3} \sin(7x) \sin(3x) + \frac{7}{9} \cos(7x) \cos(3x) \\ &+ \frac{49}{9} \int \sin(7x) \cos(3x) dx \\ &\Rightarrow \frac{40}{9} \int \sin(7x) \cos(3x) dx \\ &= \frac{1}{3} \sin(7x) \sin(3x) + \frac{7}{9} \cos(7x) \cos(3x) \\ &\Rightarrow \int \sin(7x) \cos(3x) dx \\ &= \frac{-3}{40} \sin(7x) \sin(3x) - \frac{7}{40} \cos(7x) \cos(3x) + C \end{aligned}$$

(2) (1) (3) #

$$\begin{aligned} & \int \sin(7x) \cos(3x) dx \\ &= -\frac{1}{7} \cos(3x) \cos(7x) - \frac{3}{49} \sin(3x) \sin(7x) \\ &+ \frac{9}{49} \int \sin(7x) \cos(3x) dx \\ &\Rightarrow \frac{40}{49} \int \sin(7x) \cos(3x) dx \\ &= -\frac{1}{7} \cos(3x) \cos(7x) - \frac{3}{49} \sin(3x) \sin(7x) \\ &\Rightarrow \int \sin(7x) \cos(3x) dx \\ &= \frac{-7}{40} \cos(3x) \cos(7x) - \frac{3}{40} \sin(3x) \sin(7x) + C \end{aligned}$$

(2) (1) (3) #

(法二)

$$\begin{aligned} & \int \sin(7x) \cos(3x) dx \\ &= \frac{1}{7} \cos(7x) \cos(3x) - \int \frac{3}{7} \cos(7x) \sin(3x) dx \quad (2) \\ &= \frac{1}{7} \cos(7x) \cos(3x) - \frac{3}{7} \left[\frac{1}{7} \sin(7x) \sin(3x) - \int \frac{3}{7} \sin(7x) \cos(3x) dx \right] \quad (2) \\ &= \frac{1}{7} \cos(7x) \cos(3x) - \frac{3}{49} \sin(7x) \sin(3x) + \frac{9}{49} \int \sin(7x) \cos(3x) dx \\ &\Rightarrow \int \sin(7x) \cos(3x) dx = \frac{49}{40} \left[\frac{1}{7} \cos(7x) \cos(3x) - \frac{3}{49} \sin(7x) \sin(3x) \right] + C \\ &= \frac{-7}{40} \cos(7x) \cos(3x) - \frac{3}{40} \sin(7x) \sin(3x) + C \end{aligned}$$

or (3) (1) #

$$\begin{aligned} & \int \sin(7x) \cos(3x) dx \\ &= \frac{1}{3} \sin(7x) \sin(3x) - \int \frac{7}{3} \cos(7x) \sin(3x) dx \quad (2) \\ &= \frac{1}{3} \sin(7x) \sin(3x) - \frac{7}{3} \left[-\frac{1}{3} \cos(7x) \cos(3x) - \int \frac{7}{3} \sin(7x) \cos(3x) dx \right] \quad (2) \\ &= \frac{1}{3} \sin(7x) \sin(3x) + \frac{7}{9} \cos(7x) \cos(3x) + \frac{49}{9} \int \sin(7x) \cos(3x) dx \\ &\Rightarrow \int \sin(7x) \cos(3x) dx = \frac{9}{40} \left[\frac{1}{3} \sin(7x) \sin(3x) - \frac{7}{9} \cos(7x) \cos(3x) \right] + C \\ &= \frac{-3}{40} \sin(7x) \sin(3x) - \frac{7}{40} \cos(7x) \cos(3x) + C \end{aligned}$$

(3) (1) #

(法三)

$$\begin{aligned} & \sin(10x) = \sin(7x) \cos(3x) + \cos(7x) \sin(3x) \quad (2) \\ &\Rightarrow \int \sin(7x) \cos(3x) dx \\ &= \int \sin(10x) - \cos(7x) \sin(3x) dx \\ &= \frac{1}{10} \cos(10x) - \int \cos(7x) \sin(3x) dx \\ & \int \cos(7x) \sin(3x) dx \\ &= \frac{1}{7} \sin(7x) \sin(3x) - \int \frac{3}{7} \sin(7x) \cos(3x) dx \\ &\Rightarrow \int \sin(7x) \cos(3x) dx \\ &= \frac{1}{10} \cos(10x) - \frac{1}{7} \sin(7x) \sin(3x) \\ &+ \frac{3}{7} \int \sin(7x) \cos(3x) dx \\ &\Rightarrow \int \cos(7x) \sin(3x) dx \\ &= \frac{7}{4} \left[\frac{1}{10} \cos(10x) - \frac{1}{7} \sin(7x) \sin(3x) \right] + C \\ &= \frac{7}{40} \cos(10x) - \frac{1}{4} \sin(7x) \sin(3x) + C \end{aligned}$$

(2) (1) #

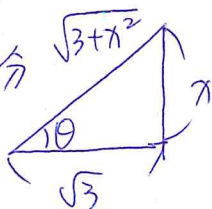
題號

1(5)

$$1(5) \int \frac{\sqrt{x^2+3}}{x} dx$$

$$\text{Let } x = \sqrt{3} \tan \theta \quad +1 \text{ 分}$$

$$\frac{dx = \sqrt{3} \sec^2 \theta d\theta}{\sec \theta = \frac{\sqrt{3}}{\sqrt{3+x^2}}} \quad +1 \text{ 分}$$



$$= \int \frac{\sqrt{3 \tan^2 \theta + 3}}{x} dx$$

$$= \int \frac{\sec \theta}{\tan \theta} \sqrt{3} \sec^2 \theta d\theta \quad +2$$

$$= \sqrt{3} \left(\int \sec^2 \theta \frac{1}{\tan \theta} \sec \theta d\theta \right)$$

$$= \sqrt{3} \int (\tan^2 \theta + 1) \frac{1}{\tan \theta} \sec \theta d\theta$$

$$= \sqrt{3} \int (\tan \theta + \cot \theta) \sec \theta d\theta$$

$$= \sqrt{3} \int \tan \theta \sec \theta d\theta + \sqrt{3} \int \cot \theta \sec \theta d\theta \quad +1$$

$$= \sqrt{3} \sec \theta + \sqrt{3} \int \csc \theta d\theta + C$$

$$\stackrel{\text{th. 1}}{=} \sqrt{3} \sec \theta - \sqrt{3} \ln |\csc \theta + \cot \theta| + C \quad +1 \quad \stackrel{\text{th. 2}}{=} \sqrt{3} \sec \theta + \sqrt{3} \ln |\csc \theta - \cot \theta|$$

$$= \sqrt{3+x^2} - \sqrt{3} \ln \left| \frac{\sqrt{3+x^2} + \sqrt{3}}{x} \right| \quad +2 \quad \cancel{\times} \quad \frac{-\sqrt{3} \ln \left| \frac{\sqrt{x^2+3}}{x} + \frac{\sqrt{3}}{x} \right|}{\quad} \quad +2$$

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題號 1. (b)

原 = $\frac{1}{2} \int \tan(2x) d(\sec(2x))$

= $\frac{1}{2} \sec(2x) \tan(2x) - \int \sec^3(2x) dx$ 全對 2分

= $\frac{1}{2} \sec(2x) \tan(2x) - \int \sec(2x) (1 + \tan^2(2x)) dx$

= $\frac{1}{2} \sec(2x) \tan(2x) - \frac{1}{2} \ln |\sec(2x) + \tan(2x)| - \text{原}$

Ans=2

$\Rightarrow \text{原} = \frac{1}{4} \sec(2x) \tan(2x) - \frac{1}{4} \ln |\sec(2x) + \tan(2x)| + C$ #

(3分)

原 = $\int \overset{①}{\sec^3(2x)} - \overset{②}{\sec(2x)} dx$ 拆項 2分

① — = $\frac{1}{2} \sec(2x) \tan(2x) - \int \sec(2x) \tan^2(2x) dx$ (原) + C 2分

② — = $\frac{1}{2} \ln |\sec(2x) + \tan(2x)| + C$ 2分

Ans=2

$\therefore \text{原} = \frac{1}{4} (\sec(2x) \tan(2x) - \ln |\sec(2x) + \tan(2x)|) + C$

* 過程中 "ln |" "無絕對值", 扣一分

* Ans 沒有寫 "C", 扣一分

* 令 "u" or 轉成 "sin, cos" 做的. 因沒人寫對, 不加以演示

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負責題號 1-7

$$\int_0^{\frac{\pi}{2}} \frac{\cos \theta}{(\sin^2 \theta + 4 \cos^2 \theta)^{\frac{3}{2}}} d\theta$$

$$\text{Let } u = \sin \theta, du = \cos \theta$$

$$= \int_0^1 (4 - 3u^2)^{-\frac{3}{2}} du \longrightarrow \begin{array}{l} \text{變數變換2分} \\ \text{(變換後的式子與範圍要正確，否則不予計分)} \end{array}$$

$$\text{Let } u = \frac{2}{\sqrt{3}} \sin t, du = \frac{2}{\sqrt{3}} \cos t dt$$

$$= \int_0^{\frac{\pi}{3}} \frac{\sqrt{3}}{12} \sec^2 t dt = \frac{\sqrt{3}}{12} \tan t \Big|_0^{\frac{\pi}{3}} \longrightarrow \begin{array}{l} \text{三角代換2分} \\ \text{(代換後的式子與範圍要正確，否則不予計分)} \end{array}$$

$$= \frac{\sqrt{3}}{12} \times \frac{\sqrt{3} u}{\sqrt{4 - 3u^2}} \Big|_0^1 \longrightarrow \text{積分化簡正確2分}$$

$$= \frac{1}{4} \square \longrightarrow \text{答案正確2分}$$

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負責題號 1-8

$$\frac{5x^2 + 20x + 6}{x^3 + 2x^2 + x} = \frac{6}{x} + \frac{-1}{x+1} + \frac{9}{(x+1)^2}$$

有計算過程且每一部分的係數正確得 4 分
每錯一個部分扣 2 分

$$\int \frac{5x^2 + 20x + 6}{x^3 + 2x^2 + x} dx$$

$$= \int \frac{6}{x} + \frac{-1}{x+1} + \frac{9}{(x+1)^2}$$

$$= 6 \ln|x| - \ln|x+1| + \frac{-9}{(x+1)} + C$$

有計算過程且每一部分的積分正確得 4 分
每錯一個部分扣 2 分
沒有加常數扣 1 分

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負責題號 2(1)

$$\begin{aligned} & \lim_{x \rightarrow 0^+} \frac{e^x - 1 - x}{x^3} \\ &= \lim_{x \rightarrow 0^+} \frac{e^x - 1}{3x^2} \quad (2 \text{ 分}) \\ &= \lim_{x \rightarrow 0^+} \frac{e^x}{6x} \quad (2 \text{ 分}) \\ &= \infty \quad (2 \text{ 分}) \quad \text{*前面都對才有這 2 分} \end{aligned}$$

1. 如果多做一次寫成 $\lim_{x \rightarrow 0^+} \frac{e^x}{6}$ ，則扣 2 分。
2. 計算過程寫對，答案才算對。
3. 需要有計算過程，否則不予計分。
4. 題目抄錯 0 分。

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負責題號 2(2)

方法一

$$\lim_{x \rightarrow 0^+} \sin(2x)^{3x}$$

$$= \lim_{x \rightarrow 0^+} e^{3x \ln \sin(2x)} \quad (2 \text{ 分})$$

$$\text{因為 } \lim_{x \rightarrow 0^+} 3x \ln \sin(2x) = \lim_{x \rightarrow 0^+} \frac{\ln \sin(2x)}{\frac{1}{3x}} = \lim_{x \rightarrow 0^+} \frac{\frac{2 \cos(2x)}{\sin(2x)}}{\frac{1}{-3x^2}} = \lim_{x \rightarrow 0^+} \frac{-6x^2 \cos(2x)}{\sin(2x)}$$

$$= \lim_{x \rightarrow 0^+} \frac{-12x \cos(2x) + 12x^2 \sin(2x)}{2 \cos(2x)} = 0 \quad (2 \text{ 分})$$

$$\text{所以 } \lim_{x \rightarrow 0^+} \sin(2x)^{3x} = \lim_{x \rightarrow 0^+} e^{3x \ln \sin(2x)} = 1 \quad (2 \text{ 分}) \quad \underline{\text{* 前面都對才有這 2 分}}$$

方法二

$$\text{假設 } y = \lim_{x \rightarrow 0^+} \sin(2x)^{3x},$$

$$\text{兩邊取 } \ln, \ln y = \lim_{x \rightarrow 0^+} \ln \sin(2x)^{3x} \quad (2 \text{ 分})$$

$$\ln y = \lim_{x \rightarrow 0^+} 3x \ln \sin(2x) = 0 \quad (\text{詳細過程請參考方法一}) \quad (2 \text{ 分})$$

$$y = \lim_{x \rightarrow 0^+} \sin(2x)^{3x} = 1 \quad (2 \text{ 分}) \quad \underline{\text{* 前面都對才有這 2 分}}$$

如果最後只寫到 $\ln y = 0$ ，扣 2 分。

1. 需要有計算過程，否則不予計分。
2. 計算過程寫對，答案才算對。

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題號 3(8分) Find the arc length of the graph $y = \frac{1}{2}(e^x + e^{-x})$, $0 \leq x \leq 2$.

sol:

$$\begin{aligned}
 \text{Arc length} &= \int_0^2 \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_0^2 \sqrt{1 + dy^2} \\
 &= \int_0^2 \sqrt{1 + \left(\frac{1}{2}(e^x - e^{-x})\right)^2} dx \quad \text{有寫出來就有4分} \\
 &= \int_0^2 \sqrt{\left(\frac{1}{2}(e^x + e^{-x})\right)^2} dx \\
 &= \int_0^2 \frac{1}{2}(e^x + e^{-x}) dx \\
 &= \frac{1}{2} (e^x - e^{-x}) \Big|_0^2 \\
 &= \frac{1}{2} (e^2 - e^{-2}) \quad \text{答案對4分}
 \end{aligned}$$

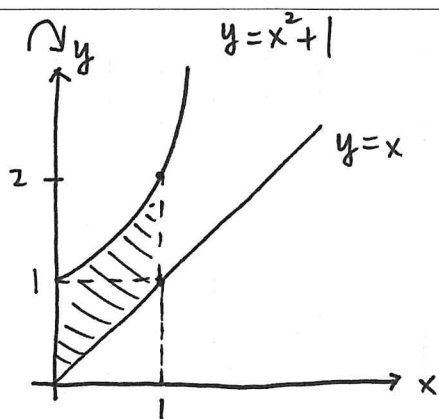
* 只要答案不對, 4分就沒了

* 公式書寫錯誤, 如多寫 π , 少寫 dx 者, 扣4分, 其餘筆誤斟酌扣分

* 計算不定積分者, 答案加 C 扣2分

* 只撰寫不定積分公式者, 該題沒分 ($\int \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$)

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題號 4 (1)

(法1)

$$\begin{aligned}
 V &= \int_0^1 2\pi \cdot x (x^2 + 1 - x) dx && (4分) \quad \star (列錯不給分) \\
 &= 2\pi \int_0^1 (x^3 - x^2 + x) dx && (1分) \\
 &= 2\pi \left(\frac{1}{4}x^4 - \frac{1}{3}x^3 + \frac{1}{2}x^2 \right) \Big|_0^1 && (1分) \\
 &= 2\pi \left(\frac{1}{4} - \frac{1}{3} + \frac{1}{2} \right) && (1分) \\
 &= \frac{5}{6}\pi && (1分)
 \end{aligned}$$

(法2)

$$\begin{aligned}
 V &= \pi \left(\int_0^1 y^2 dy + \int_1^2 (1^2 - (\sqrt{y}-1)^2) dy \right) && (4分) \quad \star (列錯不給分) \\
 &= \pi \left(\int_0^1 y^2 dy + \int_1^2 (1 - y + 1) dy \right) && (1分) \\
 &= \pi \left(\left(\frac{1}{3}y^3 \right) \Big|_0^1 + \left(-\frac{1}{2}y^2 + 2y \right) \Big|_1^2 \right) && (1分) \\
 &= \pi \left(\frac{1}{3} - 2 + 4 + \frac{1}{2} - 2 \right) && (1分) \\
 &= \frac{5}{6}\pi && (1分)
 \end{aligned}$$

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題號 4(2)

$$V = \int_0^1 \pi (x^2+2)^2 - \pi (x+1)^2 dx \quad (4分) (全列對才給分)$$

$$= \pi \int_0^1 x^4 + 3x^2 - 2x + 3 dx \quad (1分)$$

$$= \pi \left(\frac{1}{5} x^5 + x^3 - x^2 + 3x \right) \Big|_0^1 \quad (1分)$$

$$= \frac{16}{5} \pi \quad (2分)$$

$$\begin{aligned} \langle \text{力} \rangle \quad V &= 2\pi \int_0^1 (y+1) y dy + 2\pi \int_1^2 (y+1)(1-\sqrt{y-1}) dy \quad (4分) (列對才給分) \\ &= 2\pi \left[\left(\frac{y^3}{3} + \frac{y^2}{2} \right) \Big|_0^1 + \int_0^1 u+2 - u^{\frac{3}{2}} - 2u^{\frac{1}{2}} du \right] \quad (1分) \end{aligned}$$

$$\begin{pmatrix} \hat{=} u = y+1 \\ du = dy \end{pmatrix}$$

$$= 2\pi \left[\frac{5}{6} + \left(\frac{u^2}{2} + 2u - \frac{2}{5} u^{\frac{5}{2}} - \frac{4}{3} u^{\frac{3}{2}} \right) \Big|_0^1 \right] \quad (1分)$$

$$= 2\pi \times \frac{8}{5} = \frac{16}{5} \pi \quad (2分)$$